Gene Expression Networks in Prostate Cancer With and Without Obesity

Deliverable 3 — Enrichment → PPI Networks → Hubs & Communities

Course & Instructors

Course:

MO413 — Data Science & Visualization for Health (UNICAMP)

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Project Details

• Repo: **GitHub**

• Dataset: GSE79021 (FFPE)

• Tools: Orange, Cytoscape, STRING

Agenda

Prostate Cancer: Context and Risk Factors

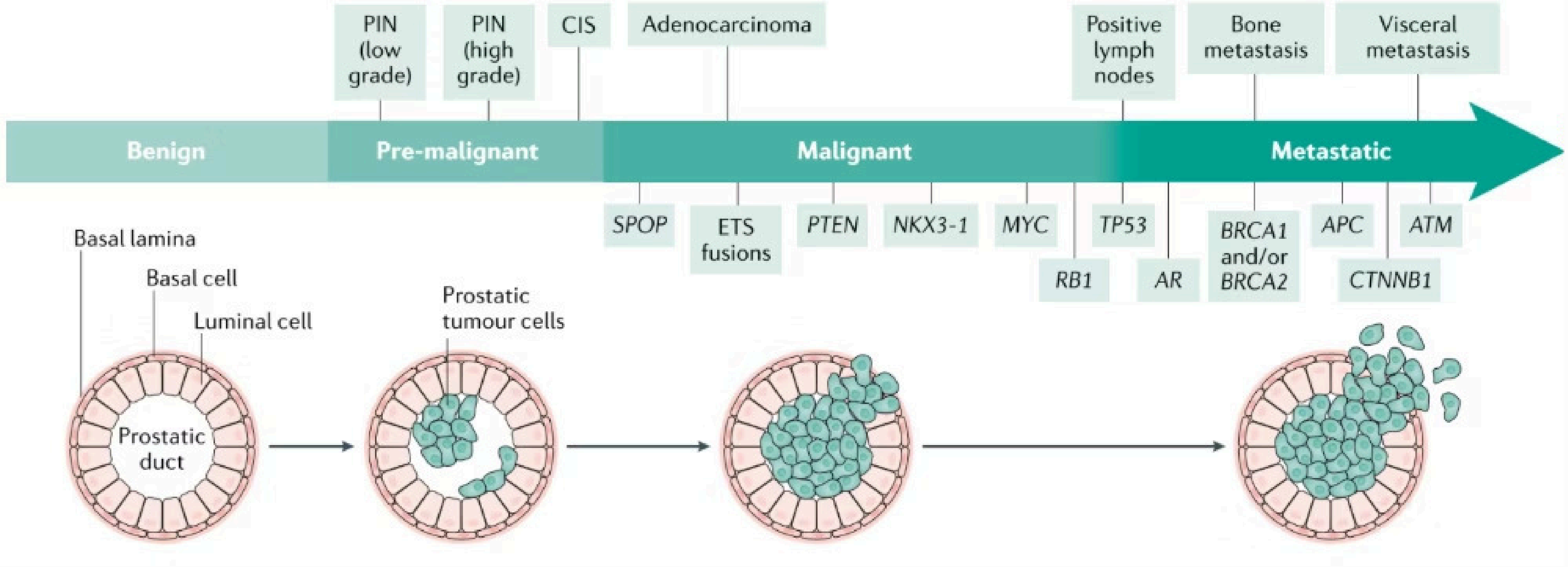
Prostate Cancer Context

- Prostate cancer (PCa) is the most incident and the deadliest cancer in men in Brazil, after non-melanoma skin cancers.
- INCA (Brazilian National Cancer Institute) data (2022): 71,730 new cases estimated for 2023-2025
- Estimated risk: ~67 new cases per 100,000 inhabitants
- Approximately 75% of new cases occur from age 65 onwards

Progression and Risk Factors

- Main risk factors: age, heredity (BRCA2, Lynch syndrome), ethnicity, exposure to carcinogens, radiation, hormonal deregulation, obesity, and diet

STAGES OF PROSTATE CANCER PROGRESSION



Obesity: Context and Epidemiology

Definition and Diagnosis

- Obesity is a metabolic condition characterized by excessive accumulation of body fat
- Multifactorial causes: genetic predisposition, endocrine disorders, inadequate eating habits, and physical inactivity
- Diagnosis: BMI (kg/m^2) - overweight: 25.0-29.9; obesity: ≥30
- BMI limitation: does not distinguish lean/fat mass nor assess fat distribution
- Visceral fat: component most associated with metabolic and oncological risks

Global and National Epidemiology

- Global estimates: >3.3 billion people affected by 2035 (increase >50% vs 2020)
- Brazil (2025): ~68% of the adult population with high BMI, 31% with obesity
- Annual growth projections: 1.9% in adults and 1.8% in children

Theoretical Foundation

nature > nature reviews urology > review articles > article

Review Article | Published: 17 May 2023

Obesity and prostate cancer – microenvironmental roles of adipose tissue

Achinto Saha, Mikhail G. Kolonin & John DiGiovanni

Nature Reviews Urology **20**, 579–596 (2023) | Cite this article

Obesity and Prostate Cancer: Evidence and Controversies

The relationship between obesity and the risk of developing prostate cancer remains controversial. Although some studies do not indicate a conclusive association, **various evidence suggests that obesity is related to an increased risk of metastatic disease and lower survival in patients with androgen-dependent prostate cancer, especially in those who experienced abrupt weight gain.**

In a cohort of 2,559 non-smokers with localized prostate cancer, it was observed that an average increase of 0.45 kg per year from age 21 until diagnosis **was associated with a higher risk of progression to the lethal form of the disease.**

Similarly, patients who gained between **9.1 and 13.6 kg had a 1.64 times higher risk of developing lethal prostate cancer,** while those who gained more than **13.6 kg had a 1.59 times higher risk, when compared to individuals with stable weight.**

Obesity and Prostate Cancer: Mortality Evidence

Overall, although results on the association between obesity and the risk of developing prostate cancer are still inconsistent, evidence linking excess weight to tumor progression and mortality is more robust. Large-scale studies demonstrate that increased BMI is positively correlated with prostate cancer-specific mortality.

In cohorts followed by the American Cancer Society, it was observed that men with grade I obesity showed a **20% increase in mortality**, while those with grade II obesity had a **34% increase**, both compared to individuals with BMI within the normal range.

Obesity: Impact on Tumor Progression and Treatment

Other studies reinforce this trend by showing that overweight and obesity are significantly associated with **prostate cancer-specific mortality** and **progression to castration-resistant forms**. It is believed that obesity favors **increased tumor grade** and **disease aggressiveness** through local and systemic metabolic and inflammatory alterations, in addition to influencing **therapeutic efficacy**.

In fact, obese patients have a **higher probability of biochemical failure** after radical prostatectomy or radiotherapy, and a **higher risk of clinical recurrence** in the form of distant metastasis.

Obesity: Quantified Clinical Impact

Studies show that each 5 kg/m² increment in BMI is associated with a **21% increase** in the risk of biochemical recurrence. Additionally, excess adipose tissue can alter the pharmacokinetics of chemotherapeutic drugs, resulting in **lower exposure of tumor cells to the drugs**.

Thus, the body of evidence indicates that, although the contribution of obesity to the initial risk of prostate cancer is still debated, its impact on disease progression, recurrence, and mortality is widely recognized.

Research Questions

Hypothesis

H1: Obesity/overweight promotes alterations in the gene expression profile in prostate tumors, resulting in the overexpression of metabolically active genes associated with parameters of tumor aggressiveness and worse clinical prognosis.

Questions

1. Does obesity alter the gene expression pattern in prostate cancer patients' tumors, resulting in a distinct set of differentially expressed genes between obese/overweight and normal-weight individuals?
2. Will the differentially expressed genes in tumors of obese/overweight patients be enriched in pathways related to inflammation, metabolism, and tumor microenvironment remodeling?
3. Will a signature score derived from the differentially expressed genes be associated with worse clinical prognosis (e.g., higher Gleason scores, more advanced stages, and shorter progression-free survival)?

Recap (Deliverable 2): Differential Expression

Dataset: **GSE79021** (FFPE; SOFT parsed in R)

Thresholds used on volcano plots: **|log2FC| ≥ 0.30** and **p < 0.05**

DEG counts

- TumorNP vs NormalNP: **10 up / 8 down**
- TumorOB vs NormalOB: **41 up / 64 down** (*Normal_OB n=3 → exploratory*)
- TumorOB vs TumorNP: **3 up / 9 down**
- NormalOB vs NormalNP: **12 up / 27 down**

Takeaway: Tumor–normal signal dominates; obesity adds subtler modulation.

Data & Groups (GSE79021)

Tumor, non-obese (Tumor_NP)

n = **143**

Tumor, obese (Tumor_OB)

n = **10**

Normal, non-obese (Normal_NP)

n = **46**

Normal, obese (Normal_OB)

n = **3** (*very small; treat with caution*)

Graph Model & Pipeline (Cytoscape)

DEG list

.soft file

Import into Orange

(stringApp)

Create control and experimental groups

4 groups

Compute Foldchange and P-Value

log2(FC) and -log10(p-value)

Compute centralities

and network stats

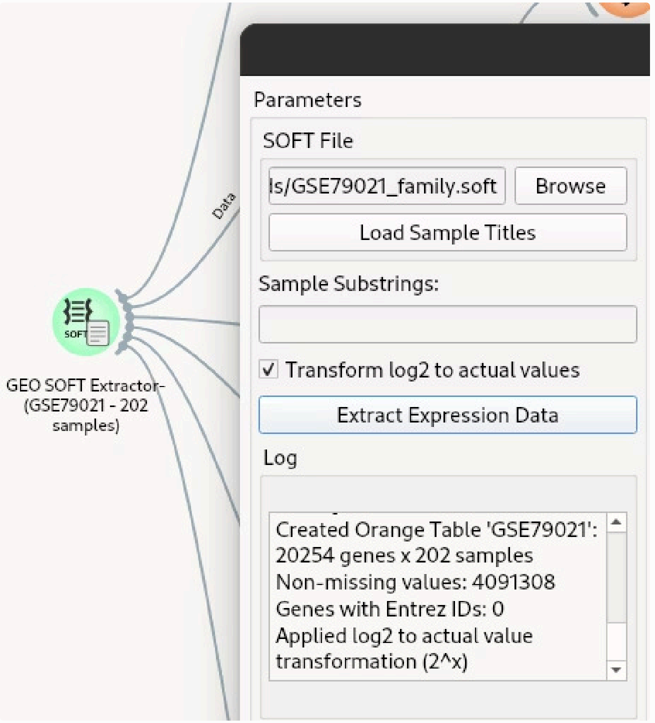
Run community detection

(Leiden/Louvain)

Export network images

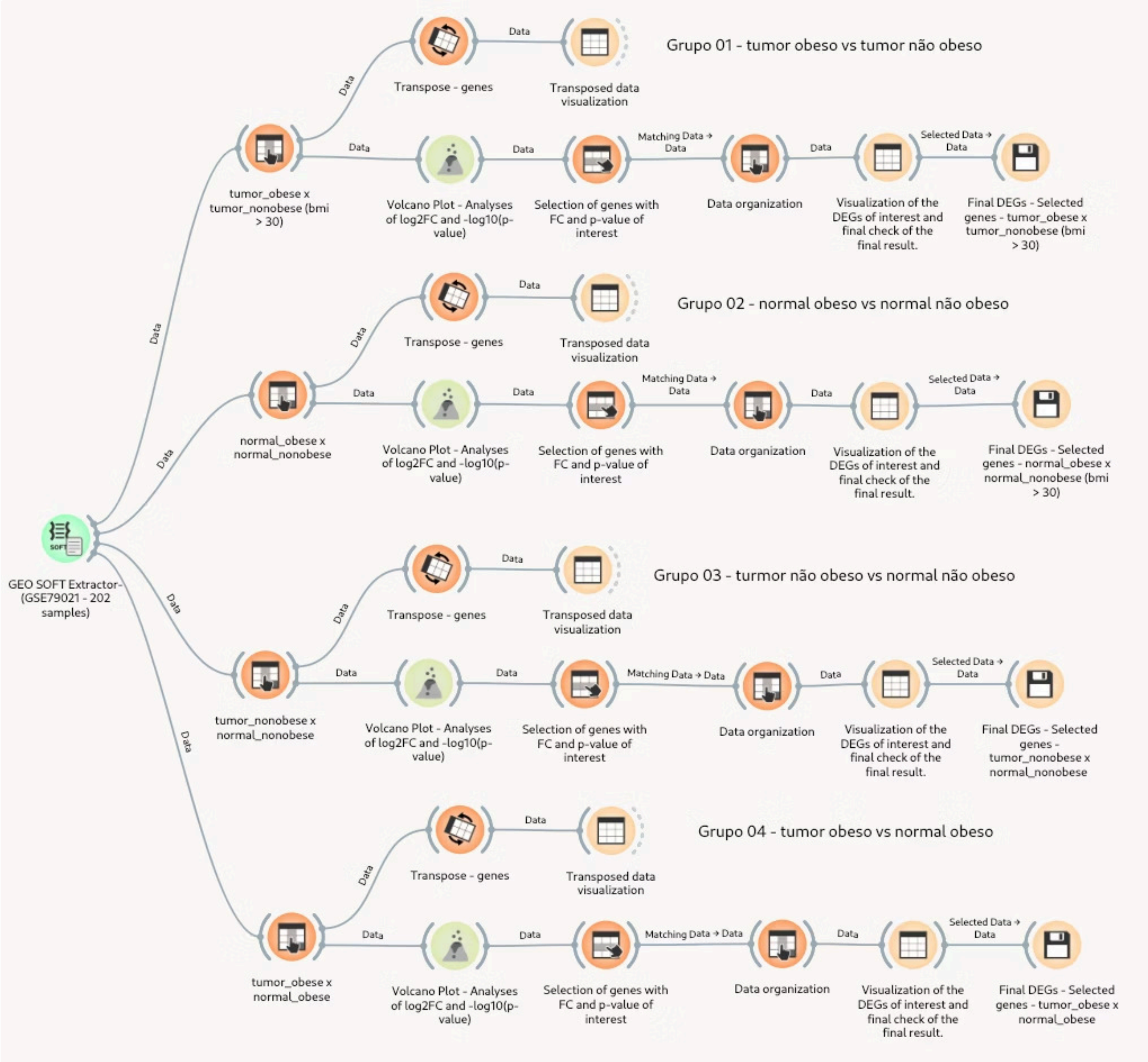
and tables for reporting

Data Extraction



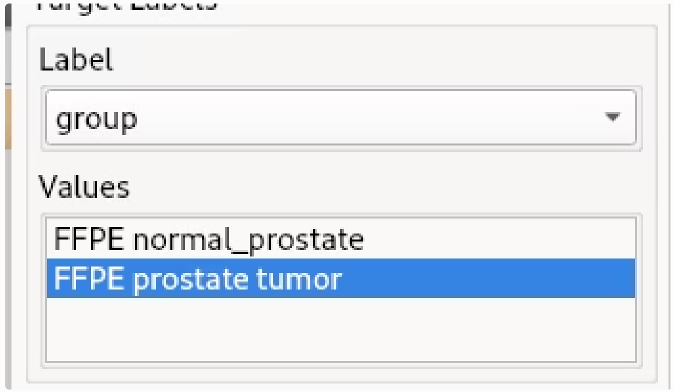
	genes	Entrez ID	Gene Symbol	GSM2083856	GSM2083857	GSM2083858	GSM2083859
bmi				22.8	24.9	29.7	24.5
class				FPE prostate t	FPE prostate t	FPE prostate t	FPE prostate t
group				PE prostate tun	PE prostate tun	PE prostate tun	PE prostate tun
1	A1BG	?	?	13.4045	14.7399	15.9206	14.9214
2	A1CF	?	?	3.07512	3.53722	4.06154	4.22466
3	A2LD1	?	?	10.4963	12.1946	13.7759	10.8492
4	A2M	?	?	65.5049	55.6715	53.7529	41.4956
5	A2ML1	?	?	4.55441	4.39214	4.77982	4.20929
6	A4GALT	?	?	7.24581	7.95212	8.23721	6.52709
7	A4GNT	?	?	3.28078	3.77793	3.83759	4.08229
8	AAA1	?	?	3.28893	3.18541	3.9365	3.41376
9	AAAS	?	?	22.5681	18.6876	16.3826	16.0603
10	AACS	?	?	12.8075	10.1619	10.8739	11.7739
11	AADAC	?	?	2.00867	2.22365	2.54854	2.00161
12	AADACL2	?	?	2.3678	2.66464	3.20168	2.83317
13	AADACL3	?	?	4.0994	3.11828	3.86494	4.80196
14	AADACL4	?	?	2.2588	2.50167	3.02727	2.7861
15	AADAT	?	?	23.3103	27.861	16.2983	18.9009
16	AAGAB	?	?	10.4962	10.1574	15.033	10.4947
17	AAK1	?	?	35.5172	55.9108	29.1379	42.7706
18	AAMP	?	?	17.5204	14.565	18.6765	14.8258
19	AANAT	?	?	7.76648	9.07235	11.7782	8.15436
20	AARS	?	?	28.7544	25.3983	22.138	45.2891
21	AARS2	?	?	5.11104	8.74947	5.35778	5.96726

Data Extraction



Data Extraction

	Feature name	bmi	class	group	A1BG	A1CF	A2LD1
1	GSM20840...	23.10000	23.1|FFPE n...	FFPE norm...	18.4632	5.71931	13.9527
2	GSM2084010	24.10000	24.1|FFPE n...	FFPE norm...	20.401	4.13814	13.1576
3	GSM2084011	36.80000	36.8|FFPE ...	FFPE norm...	18.5634	4.5798	12.5918
4	GSM2084012	29.00000	29|FFPE no...	FFPE norm...	15.7081	4.04888	11.9798
5	GSM2084013	23.10000	23.1|FFPE n...	FFPE norm...	15.2641	4.01849	13.791
6	GSM2084014	23.00000	23|FFPE no...	FFPE norm...	23.2056	3.32371	11.562
7	GSM2084015	29.00000	29|FFPE no...	FFPE norm...	15.9291	4.74788	14.0268
8	GSM2084016	22.50000	22.5|FFPE ...	FFPE norm...	15.0494	4.28444	13.9398
9	GSM2084017	24.40000	24.4|FFPE ...	FFPE norm...	12.8662	3.39238	11.6812
10	GSM2084018	25.10000	25.1|FFPE n...	FFPE norm...	17.1532	5.45961	13.4787
11	GSM2084019	24.50000	24.5|FFPE ...	FFPE norm...	14.6286	4.18372	10.918
12	GSM2084020	26.60000	26.6|FFPE ...	FFPE norm...	17.3057	3.68444	10.2322
13	GSM2084021	28.40000	28.4|FFPE ...	FFPE norm...	10.983	4.01263	9.74984
14	GSM2084022	25.80000	25.8|FFPE ...	FFPE norm...	14.8229	3.5998	12.4647
15	GSM2084023	25.70000	25.7|FFPE n...	FFPE norm...	15.1875	3.89902	14.7124
16	GSM2084024	28.80000	28.8|FFPE ...	FFPE norm...	16.3706	3.02608	12.2606
17	GSM2084025	24.50000	24.5|FFPE ...	FFPE norm...	17.357	3.43646	13.0037
18	GSM2084026	23.10000	23.1|FFPE n...	FFPE norm...	15.4782	3.32468	15.3605
19	GSM2084027	23.40000	23.4|FFPE ...	FFPE norm...	17.5563	4.05427	14.2875
20	GSM2084028	24.70000	24.7|FFPE ...	FFPE norm...	17.1232	3.81732	12.3141
21	GSM2084029	23.00000	23|FFPE no...	FFPE norm...	19.8182	3.87144	13.9134
22	GSM20840...	26.60000	26.6|FFPE ...	FFPE norm...	15.389	3.93132	16.5818



Data Extraction

-log10 (P_value)	is at least			1,300000	×
log2 (ratio)	is outside	-0,660000	and	0,660000	×

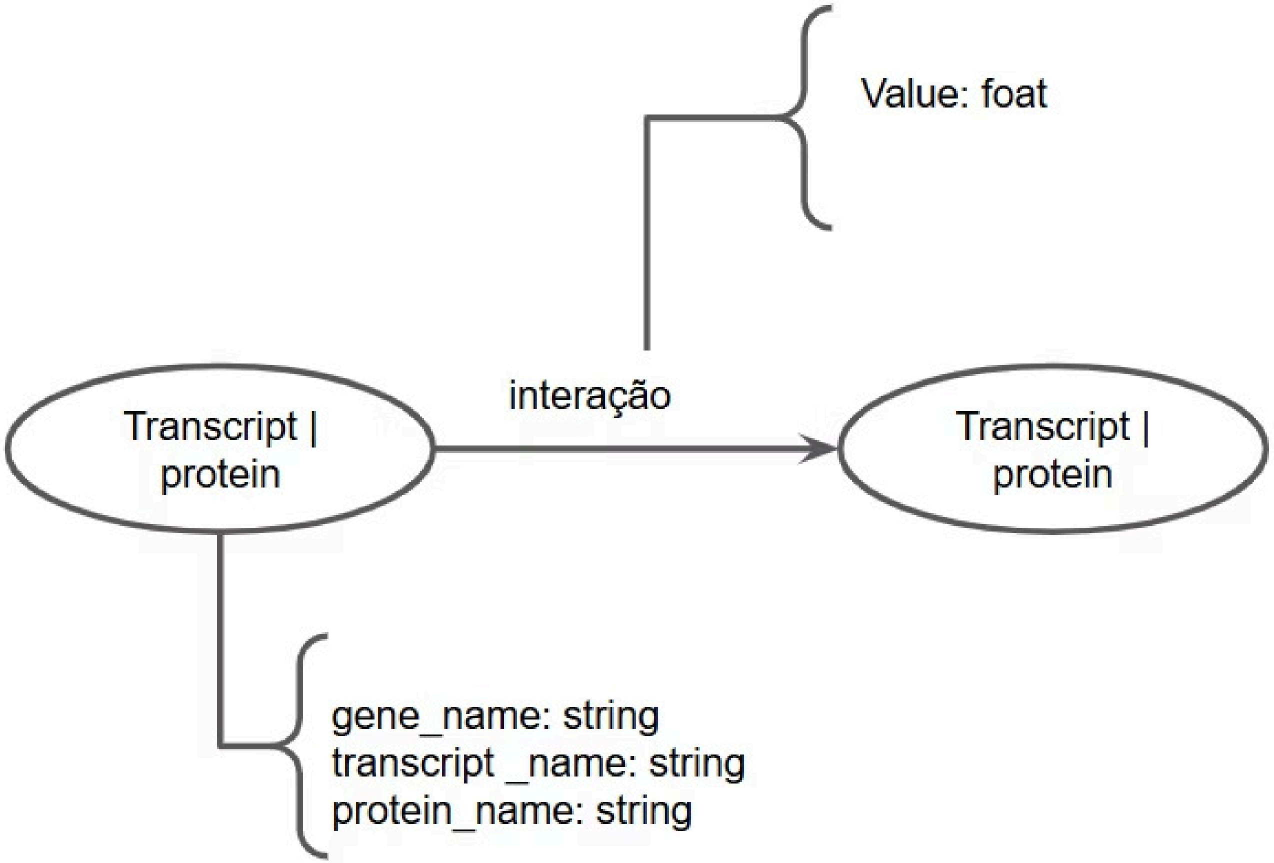
genes	log10 (P_value	log2 (ratio)
AQP7	2.82611	0.6885
B3GAT3	1.84021	0.715004
B3GNT5	2.6786	1.32336
C21orf94	4.61571	1.30925
C6orf155	3.54413	0.815686
CSGALNACT1	2.62929	0.813511
CTAGE4	2.72753	0.798467
CXCL2	1.36822	0.767078
CYB5R1	3.57437	0.798228
DSERG1	2.30855	1.04543
EMP1	1.52963	0.943411
FAM177B	1.41227	0.701035
FLJ39632	1.48512	0.842512
GAFA3	2.30059	0.756381

Data Extraction

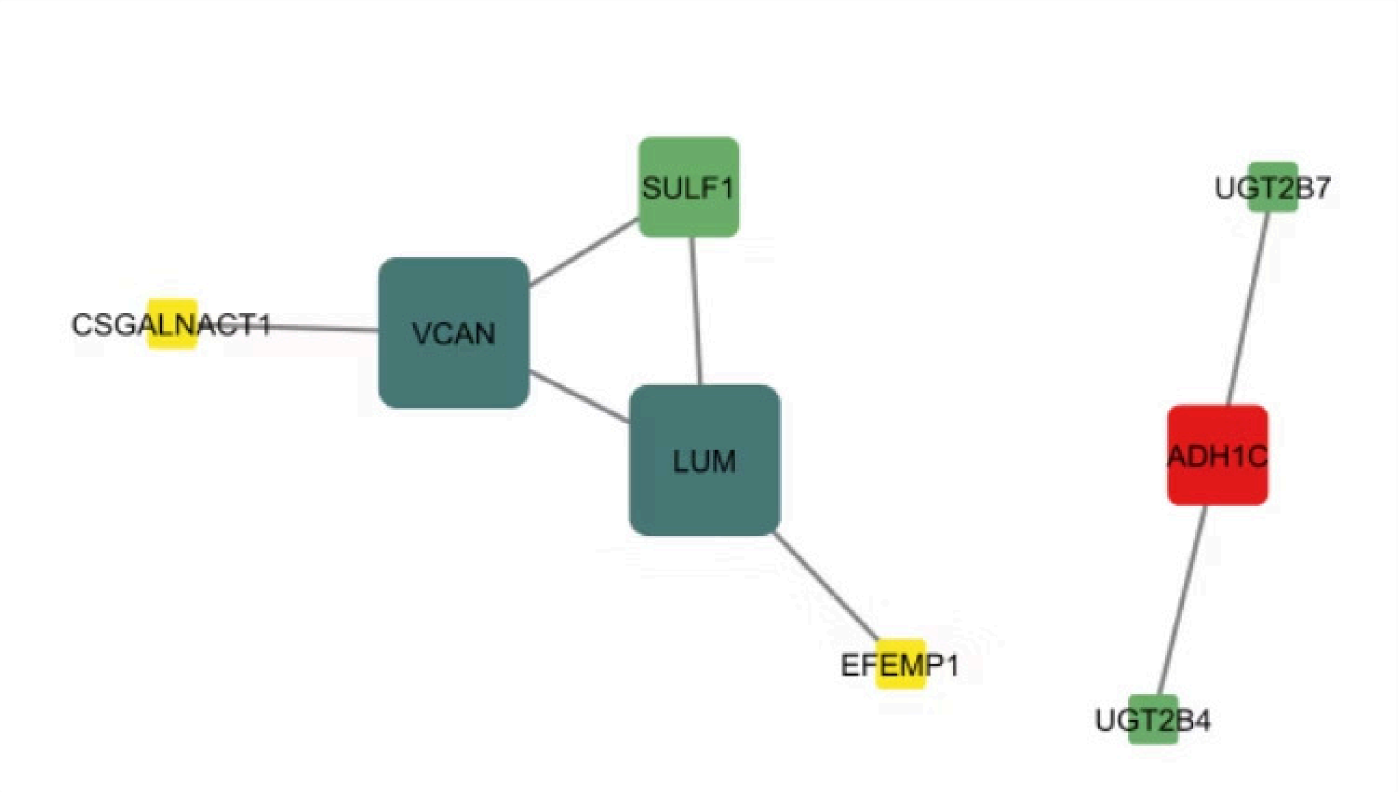
Control Group x Experimental Group	Genes Quantity
Normal não obeso x Normal obeso	49
Tumoral não obeso x Tumoral obeso	153
Normal não obeso x Tumoral não obeso	189
Normal obeso x Tumoral obeso	13

Data Extraction

Control Group x Experimental Group	Genes Quantity
Normal não obeso x Normal obeso	88
Tumoral não obeso x Tumoral obeso	38
Normal não obeso x Tumoral não obeso	56
Normal obeso x Tumoral obeso	583

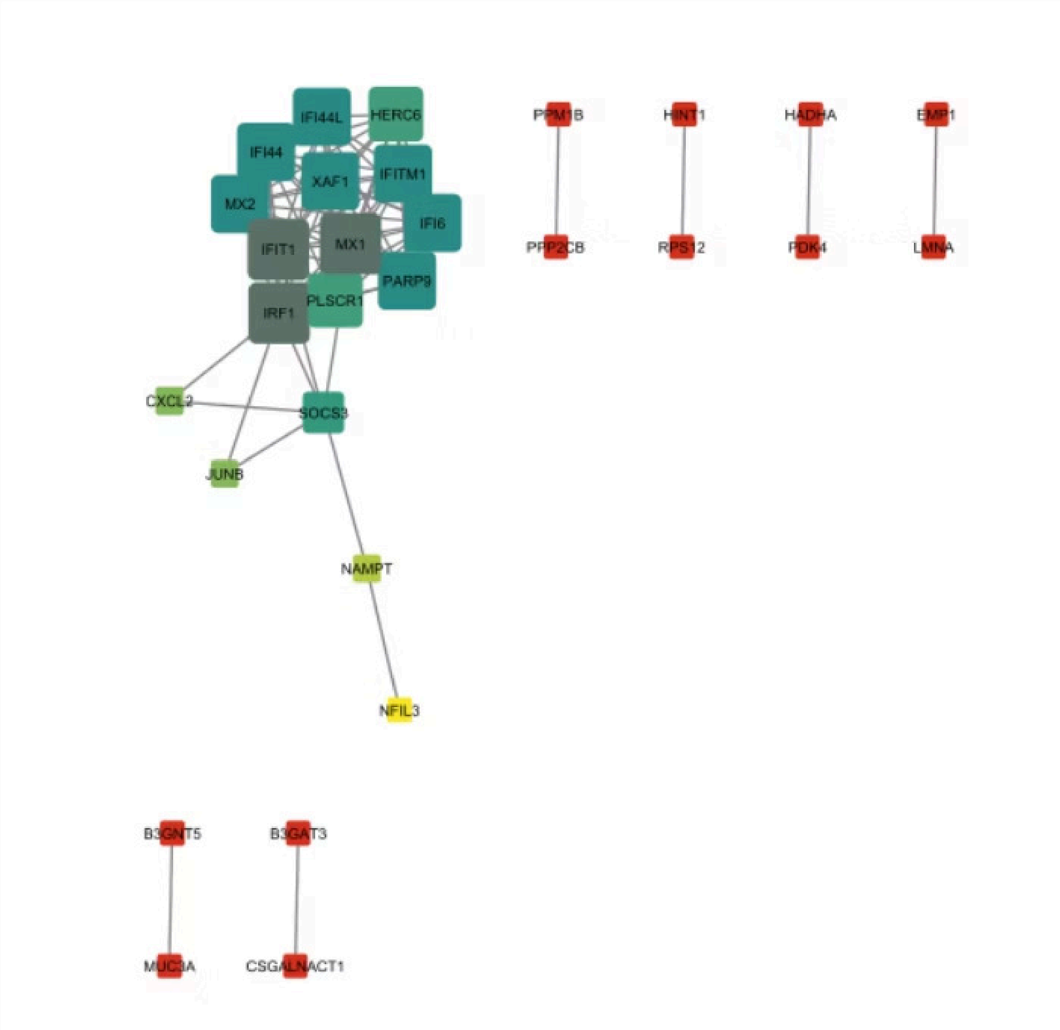

Value: foat
interação
Transcript | protein
Transcript | protein
gene_name: string
transcript _name: string
protein_name: string

Results — Gene interaction networks in analyzed subgroups



Graph generated in Cytoscape: Obese tumor vs. non-obese tumor

Results — Gene interaction networks in analyzed subgroups



Graph generated in Cytoscape: Normal obese vs normal non-obese

Results — Enrichment

Functional annotation Clustering result

Functional Annotation Clustering

Current Gene List: List_1
Current Population: Homo sapiens

Switch to Classic Version

Show options | **Classification Stringency:** Highest | **EASE Threshold:** 1 | Update Table

Create sublist | Copy | Excel | CSV | PDF | Column visibility ▾ | Search:

Sublist	Category	Term	RT	Genes	Count	%	P-Value	Benjamini
Term Cluster 1		**Enrichment Score: 2.09**		G				
☐	KEGG_PATHWAY	Retinol metabolism	RT	8.33%	3	8.33	6.99e-3	1.34e-1
☐	KEGG_PATHWAY	Drug metabolism - cytochrome P450	RT	8.33%	3	8.33	8.01e-3	1.34e-1
☐	KEGG_PATHWAY	Metabolism of xenobiotics by cytochrome P450	RT	8.33%	3	8.33	9.34e-3	1.34e-1

Common genes: **UGT2B4, UGT2B7, ADH1C**

DAVID Gene Report

Current Gene List: List_1
Current Population: Homo sapiens

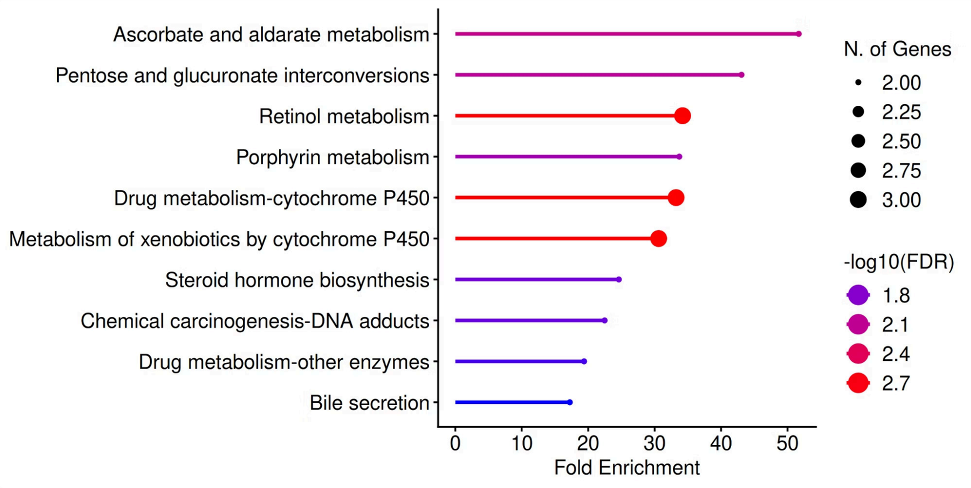
Switch to Classic Version

Copy | Excel | CSV | PDF | Search:

User ID	Gene Name	Related Genes	Species
UGT2B4	UDP glucuronosyltransferase family 2 member B4(UGT2B4)	RG	Homo sapiens
UGT2B7	UDP glucuronosyltransferase family 2 member B7(UGT2B7)	RG	Homo sapiens
ADH1C	alcohol dehydrogenase 1C (class I), gamma polypeptide(ADH1C)	RG	Homo sapiens

Top pathways

- Metabolism of xenobiotics by cytochrome P450
- Drug metabolism – cytochrome P450
- Retinol metabolism



Results — Enrichment

Metabolism of xenobiotics by cytochrome P450

Controlled by enzymes called CYPs. In prostate cancer, alterations in these enzymes can influence tumor development and response to treatment.

Drug metabolism – cytochrome P450

It focuses on drug metabolism. If CYP enzymes are too active, they can break down drugs too quickly, making treatment less effective. If they are underactive, the drugs may not work properly.

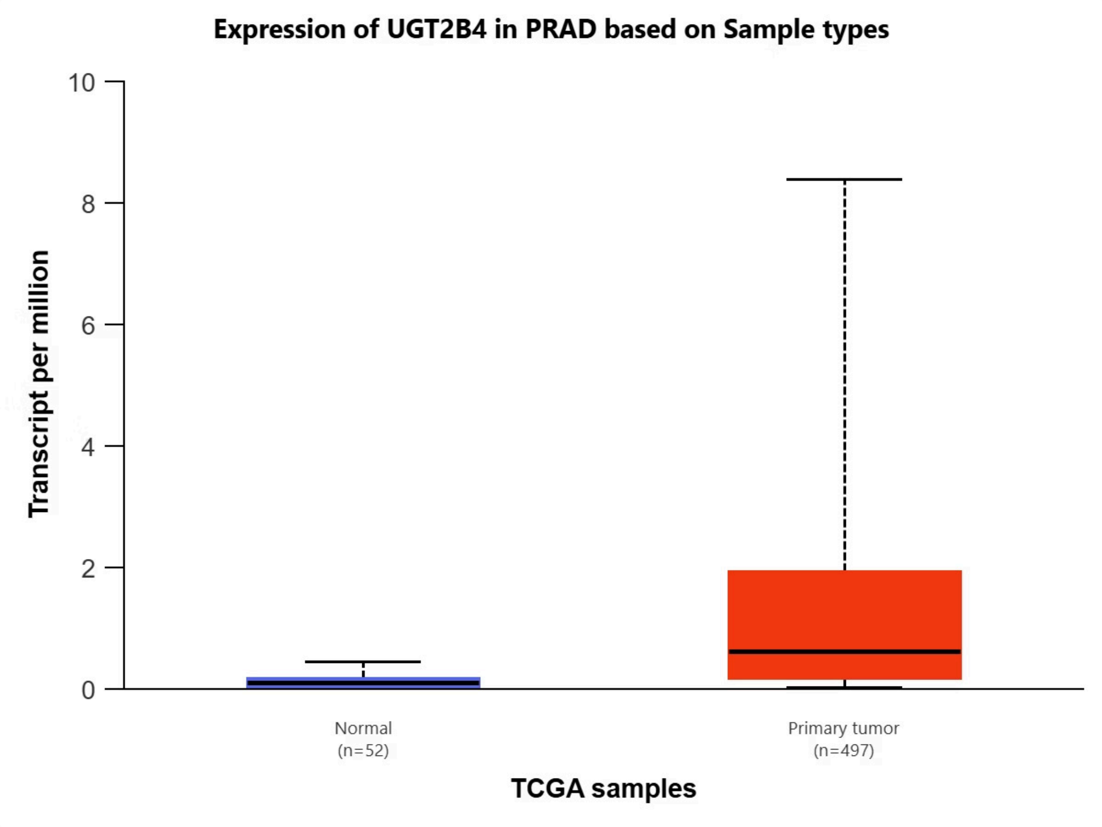
Retinol metabolism

Retinol (vitamin A) is important for cell differentiation and growth. In prostate cancer, retinol is transported from fat (adipocytes) to tumor cells. In obese individuals, because there is more fat around the prostate, this transport may be more intense, favoring tumor plasticity.

In general, across all three pathways, increased inflammation and metabolic changes resulting from obesity can dysregulate these pathways, affecting the effectiveness and resistance to chemotherapeutic agents.

Top pathways shares the exact common genes: **UGT2B4, UGT2B7, ADH1C.**

Results — Validation Analysis in Public Databases



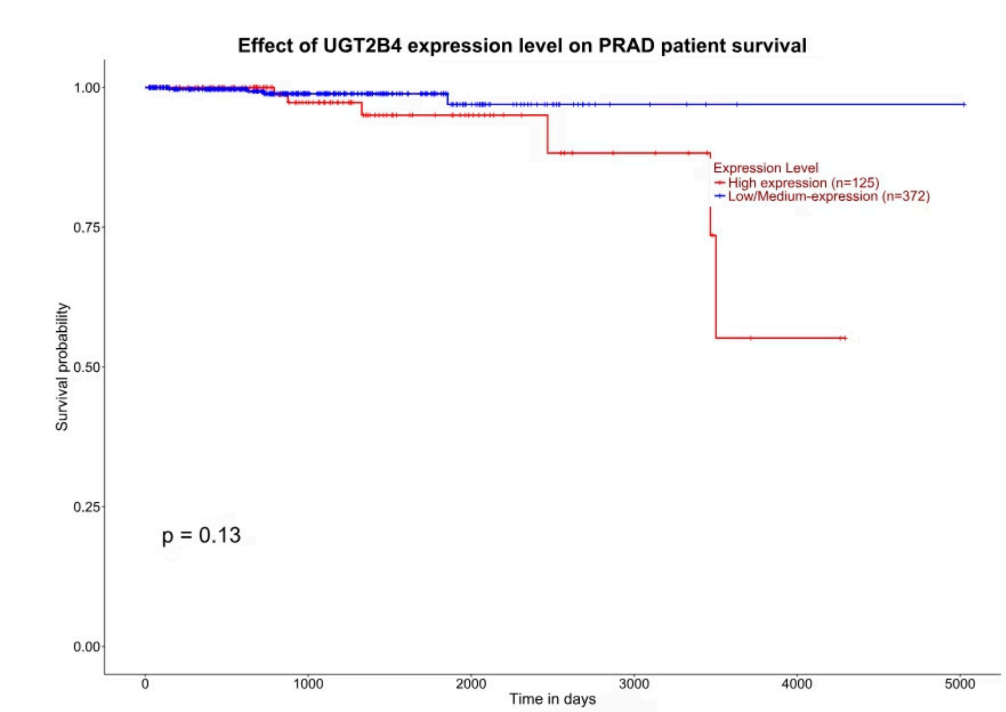
To validate the expression pattern of these genes in the context of prostate cancer, the UALCAN platform was used, which provides TCGA transcriptomics data. Analysis of the expression profile of UGT2B7 and ADH1C in the PRAD (Prostate Adenocarcinoma) dataset showed that these genes are, in fact, **downregulated in tumor tissue compared to normal tissue**, a pattern inconsistent with a potential oncogene, making them of little interest for the current analysis.

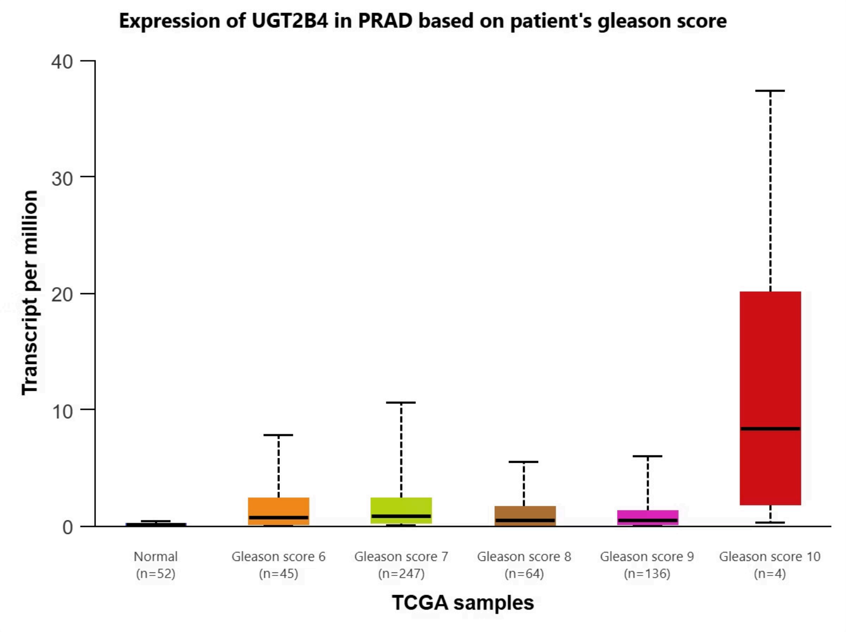
Given this incongruence, the focus of the investigation was directed to UGT2B4, which, through the same analysis in UALCAN, demonstrated a **clear significant upregulation in primary tumor tissue**, corroborating our initial dataset and aligning with a possible tumor-promoting role.

UGT2B4: Clinical Characterization and Prognosis

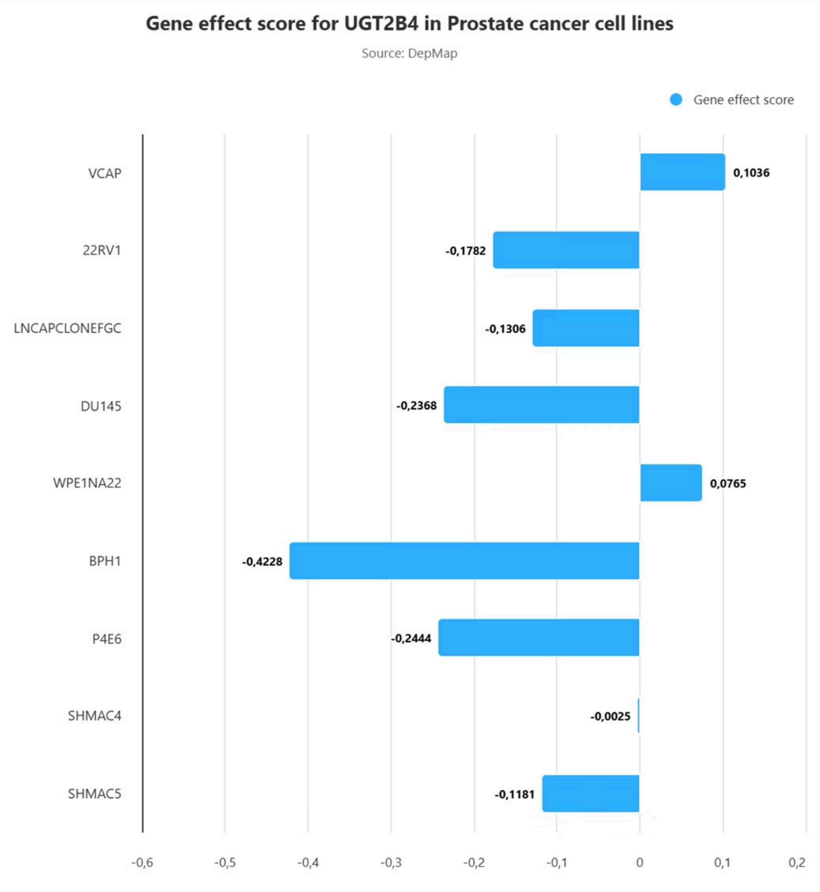
Deepening the characterization of UGT2B4, TCGA data via UALCAN further revealed that its expression correlates positively with tumor aggressiveness, being notably higher in patients with higher Gleason scores. From a clinical perspective, this high expression was shown to be associated with worse prognosis, with an overall survival analysis indicating a Hazard Ratio (HR) greater than 1.

UGT2B4: Clinical Characterization and Prognosis





UGT2B4 as a Therapeutic Target: DepMap Analysis



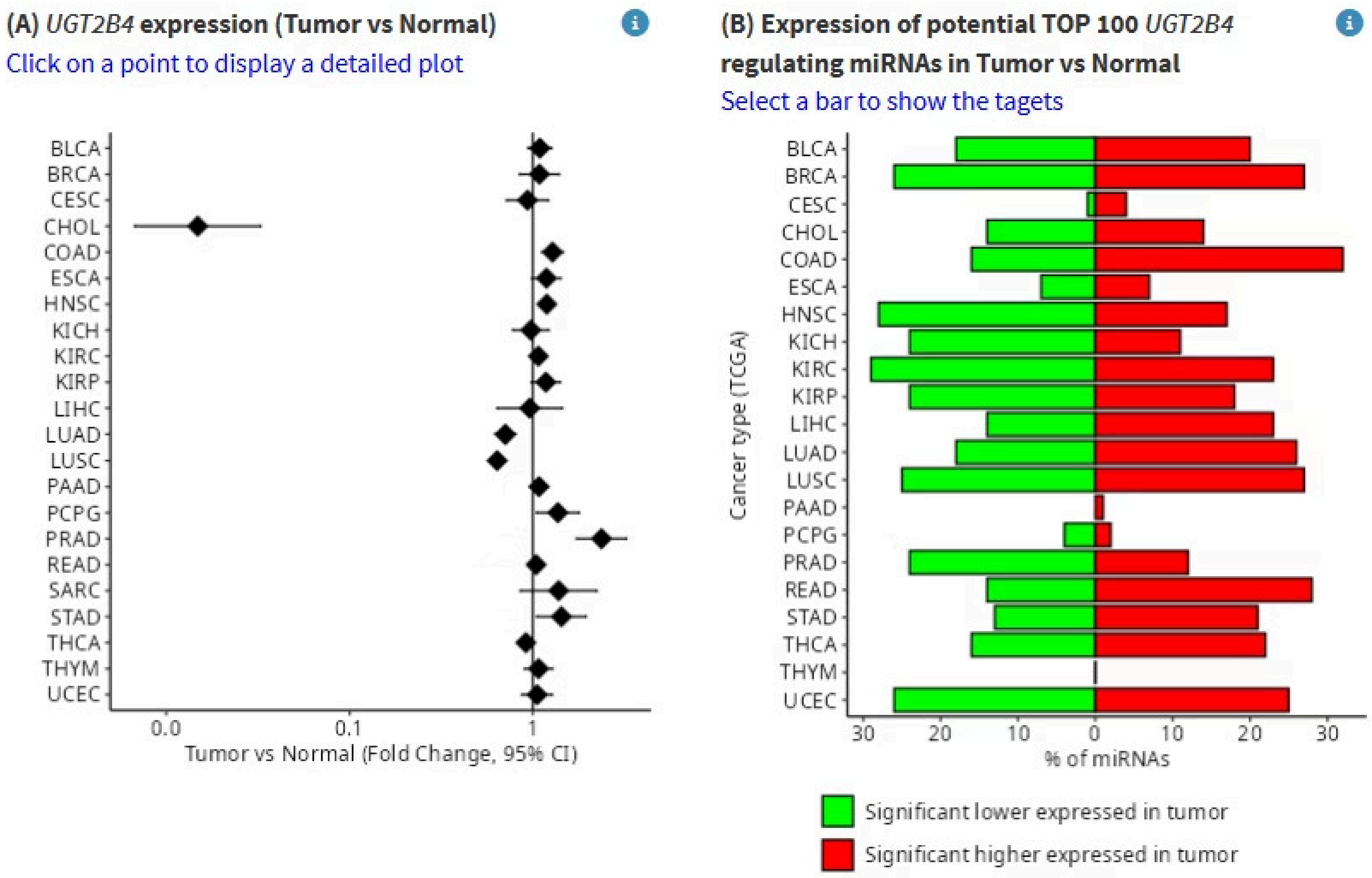
To explore the potential of UGT2B4 as a therapeutic target, public cellular dependency data generated by the DepMap project were consulted, which derive from high-throughput CRISPR knockout experiments. The interpretation of "Gene Effect scores" indicates that negative values mean that gene inactivation compromises cellular viability.

In this context, most prostate cancer cell lines analyzed showed negative scores for UGT2B4. Specifically, widely used cell lines such as **DU145**, **22RV1**, **LNCaP**, and the benign hyperplasia line **BPH-1** exhibited the most negative scores, suggesting a strong functional dependence on UGT2B4 activity.

These data from independent consortia support the hypothesis that pharmacological inhibition of UGT2B4 could be a viable therapeutic strategy for prostate cancer subtypes.

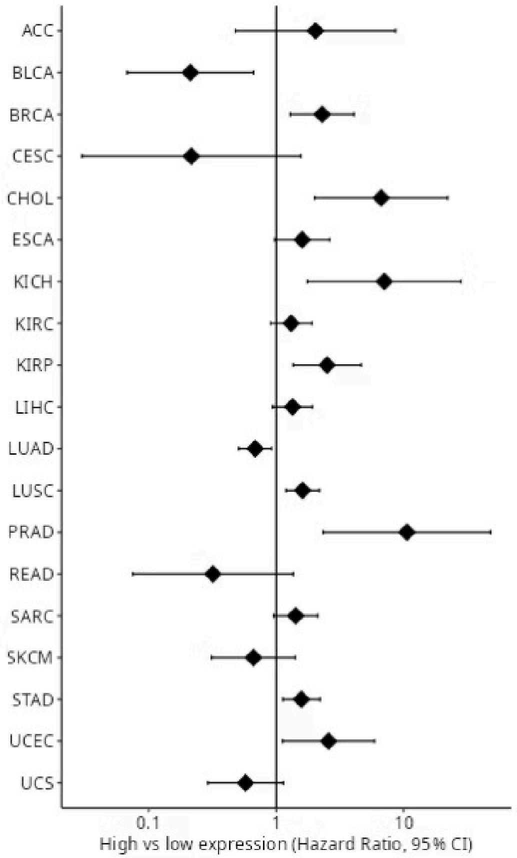
UGT2B4 Regulation: miRNA Analysis

To understand the mechanisms that regulate UGT2B4 expression, an in silico analysis of miRNA regulation was conducted using the mirtarget tool. The results indicated that **miRNAs that repress UGT2B4 are significantly less expressed in tumors**. This suppression of inhibitory miRNAs effectively removes a post-transcriptional control mechanism, allowing **UGT2B4 accumulation**. This finding provides a plausible mechanistic explanation for the observed upregulation.


(A) UGT2B4 expression (Tumor vs Normal)
Click on a point to display a detailed plot
BLCA
BRCA
CESC
CHOL
COAD
ESCA
HNSC
KICH
KIRC
KIRP
LIHC
LUAD
LUSC
PAAD
PCPG
PRAD
READ
SARC
STAD
THCA
THYM
UCEC
0.0
0.1
1
Tumor vs Normal (Fold Change, 95% CI)
(B) Expression of potential TOP 100 UGT2B4 regulating miRNAs in Tumor vs Normal
Select a bar to show the tagets
Cancer type (TCGA)
BLCA
BRCA
CESC
CHOL
COAD
ESCA
HNSC
KICH
KIRC
KIRP
LIHC
LUAD
LUSC
PAAD
PCPG
PRAD
READ
STAD
THCA
THYM
UCEC
30
20
10
0
10
20
30
% of miRNAs
Significant lower expressed in tumor
Significant higher expressed in tumor

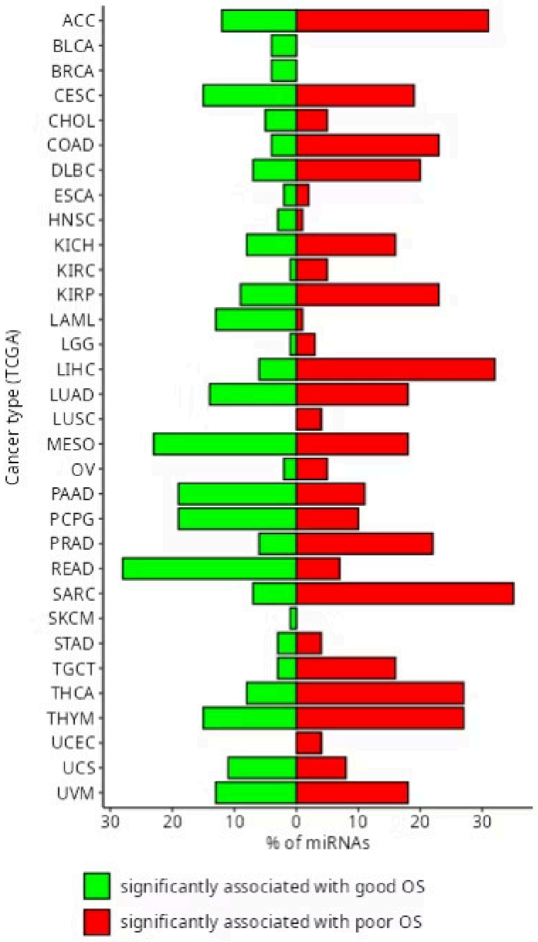
(C) Association of *UGT2B4* with overall survival

Click on a point to display a Kaplan Meier curve



(D) Association of potential TOP 100 *UGT2B4* regulating miRNAs with overall survival

Select a bar to show the miRNAs



CONCLUSION

Future Work

Expansion of Bioinformatics Study

Targeted Experimental Validation

Exploration of Mechanisms

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Thank you!

Any questions?